Planning And Design (Section2)

Koketso Rabothata-ST10447912

Lindo Mphenyeke-ST10446906

Given Mudau-ST10451061

Alan Chauke-ST10445966

Lwanda Godlwana-ST10450796

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1. Introduction

The entire blueprint for Spendwise, a mobile budgeting tool designed for contemporary South African and international consumers, is presented in this design document. It immediately expands on results from the Spendwise Research Document, which identified shortcomings in current apps, such as cluttered interfaces that cause user fatigue (common in traditional banking apps), poor trend visualisation, low retention because of low interest, and inadequate gamification.

Spendwise creates an interesting, glanceable experience by combining Fortune City's gamified features, Emma's smooth automation, and Mint's strong data structure and expert tools. The design places a strong emphasis on interactive visualisations for fast insights, behavioural cues through progress monitoring, and a high contrast dark mode for less eye strain. Every option is based on research to enhance long-term use and financial knowledge. This document connects requirements, UI flows, mockups, and a timed implementation plan with empirical insights.

2. Overiew Of The App

App Name: SpendWise

The Spendwise app icon features a classic burlap money sack tied with rope, prominently embroidered with a bold emerald-green “R” (symbolizing the South African Rand). The sack rests in a softly lit vault setting surrounded by scattered gold coins, evoking security, accumulated wealth and financial growth. The emerald-green “R” provides strong visual pop against the neutral burlap and ties directly to the app’s income and net-balance accents (emerald-green for positive flows) while the overall earthy tones ensure excellent contrast and recognizability on both dark and light home screens. This design choice reinforces Spendwise’s core purpose helping users in South Africa and beyond securely manage, visualize and grow their Rand-based finances while remaining simple, scalable and instantly communicative at small icon sizes.

Vision and Identity: With an optional persistent dark mode that highlights income (green) and expenses (red), Spendwise offers a smart, data-rich experience that enables "glance-based budgeting".

Core Goal: Act as a focal point for proactive budget changes, spending trends over a six-month period and net balance overviews providing macro-level visibility lacking in many competitors.

Innovative Features:

- ✧ Interactive doughnut chart for category proportions with tap-to-reveal amounts.
- ✧ 6-month predictive trend line graph for seasonal pattern detection.
- ✧ Gamified targets with progress bars, haptic feedback, and urgency via countdowns.
- ✧ Custom categories, receipt photo storage, CSV export, and multi-currency (default ZAR).

3. **Detailed List of Requirements**

The following requirements define the functional and performance benchmarks for Spendwise. These have been specifically expanded to support the **"Light Mode" visual architecture** and the specialized data visualization widgets identified in the initial design phase.

3.1 Functional Requirements

3.1.1 Advanced User Authentication & Profile Customization

- **Secure Registration & Onboarding:** The system must provide a "Create Account" interface (as seen in the mockups) capturing First Name, Last Name, Gender, and Email.
- **Profile Management:** Users must be able to "Change Photo," update personal details, and modify their "New Password" through a secure settings pane.
- **Security & Data Privacy:** To align with the security standards of **Mint**, the app must include a "Security & Privacy" card allowing users to "Export Data to CSV" or "Clear All Data" for GDPR compliance learn more .

3.1.2 Real-Time Financial Dashboard & Net Balance Logic

- **Net Balance Calculation:** The system must automatically calculate the "Net Balance" by subtracting total "Expenses" from "Income." This must be displayed prominently in the central dashboard card.
- **6-Month Trend Visualization:** The app must include a coordinate-based line graph tracking net balance over time. The system must support data points ranging from -0.9 to 0.9 on the Y-axis to visualize growth and deficits accurately.
- **Interactive Expenditure Doughnut:** Expenditure must be visualized through a central doughnut chart. The system must support "Category Hover" logic, where clicking a segment (e.g., "Food") displays the specific currency amount (e.g., "$20,000") in the center of the ring.

3.1.3 Granular Transaction & Budget Management

- **Transaction Ledger:** The app must maintain a chronological list of "All Transactions." Each entry must include a Directional Icon (Up/Green for Income, Down/Red for Expenses), Category Label, Date, and a "Delete" (trash can) function.
- **Flexible Budget Setting:** Users must be able to set a "Monthly Budget" via a dedicated input field. The system must calculate the "Percentage Used" and display it via a progress bar (e.g., "2% used").

3.1.4 Goal Tracking & "Budget Alerts"

- **Progressive Savings Goals:** Following the gamification research from **Fortune City**, users can set specific goals (e.g., "Emergency Fund," "Vacation"). Each goal must track:
    - Current vs. Target amount.
    - Percentage achieved (e.g., "46.0% achieved").
    - A countdown of "Days Left" until the set deadline.
- **Contextual Alerts:** The dashboard must feature a "Budget Alert" banner that triggers when spending in a specific category (e.g., Healthcare) reaching a predefined threshold.

3.2 Non-Functional Requirements

3.2.1 High-Contrast UI Performance

✧ **Light Mode Persistence:** The application's "Appearance" settings must support a global "Light Mode" toggle. This state must be saved in the user's local cache so that the app consistently launches in the user's preferred theme.

✧ **Visual Hierarchy & Haptics:** To prevent cognitive overload, the UI must use a "Card-Based" layout. Success-critical actions (like "Save Changes" or "Set Budget") must use a Coral-Red primary color to stand out against the dark background.

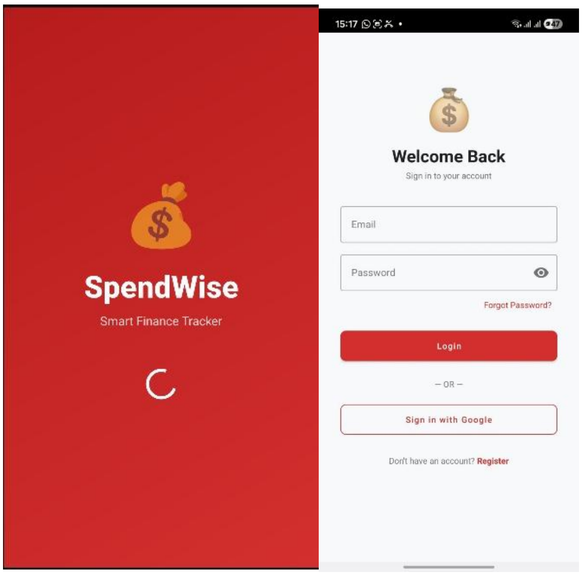
3.2.2 Data Integrity & Export

✧ **CSV Export Functionality:** The system must allow users to export their transaction history into a standard CSV format for use in external professional software like Excel, mirroring the "Professional Standard" identified in the research of **Mint**.

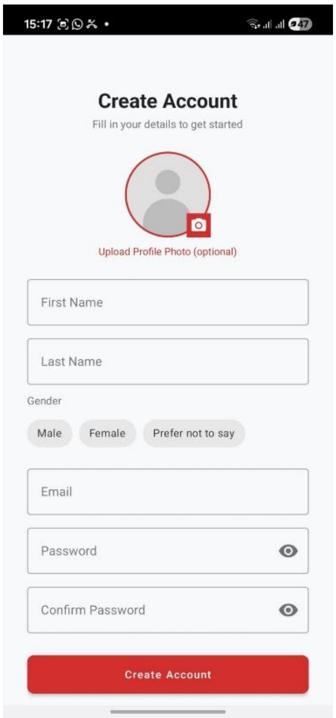
✧ **Global Currency Support:** The system must support a "Currency" selector (e.g., USD, EUR, ZAR) that automatically updates all labels across the Dashboard, Transactions, and Goals screens.

4. User Interface Design

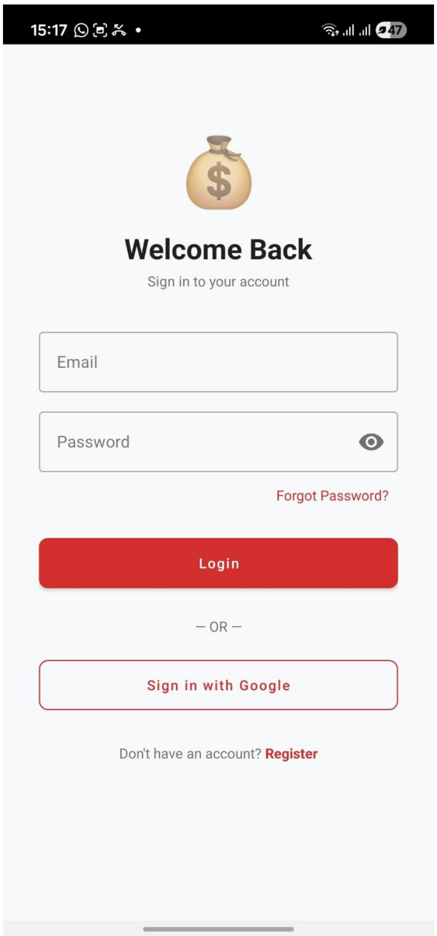
1. **Splash/Login Screen** → Simple welcome with the money bag logo, app name, login fields (email/password), google sign in method.



2. **Create Account / Sign Up Screen** → Form-focused registration with profile photo upload, name fields, gender selection, email, password/confirm, and a prominent red "Create Account" button.

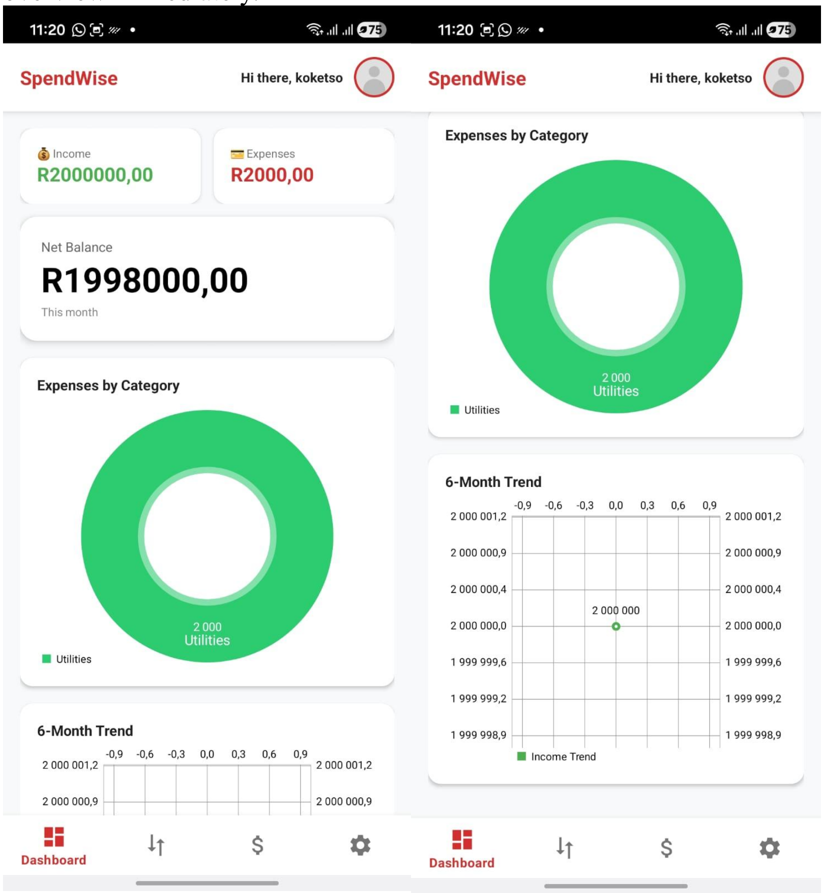


3. **Login / Sign In Screen →** Welcoming return screen with email + password fields, Google sign-in option, "Forgot Password?", and links to register.

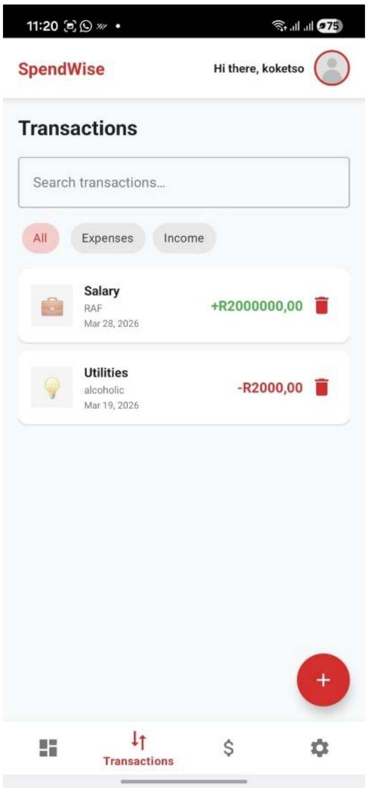


4. **Main home**: Greeting; cards for Income/Expenses/Net Balance; donut chart (Expenses by

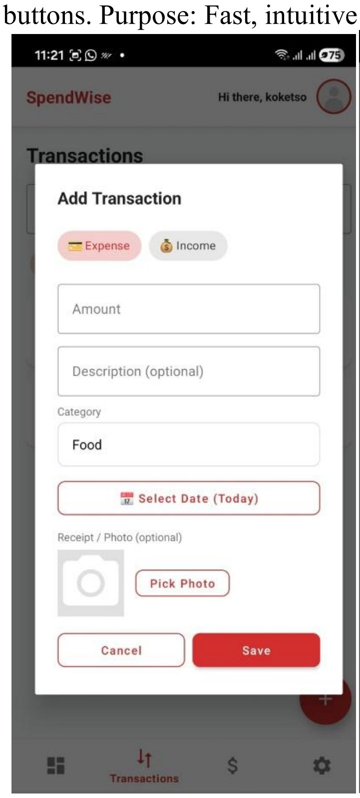
Category, e.g., green Utilities slice); 6-month trend line chart. Purpose: Quick financial overview immediately.

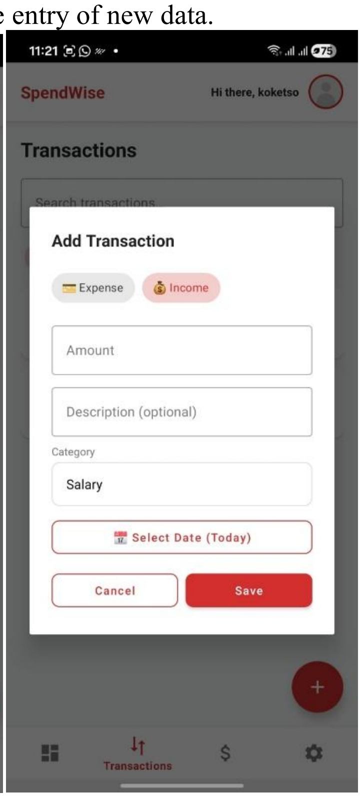


5. **Transactions Screen** → List of entries (cards with category icon, description, amount/date, delete); search bar; tabs (All/Expenses/Income); floating + button to add. Purpose: View, search, manage all money movements.

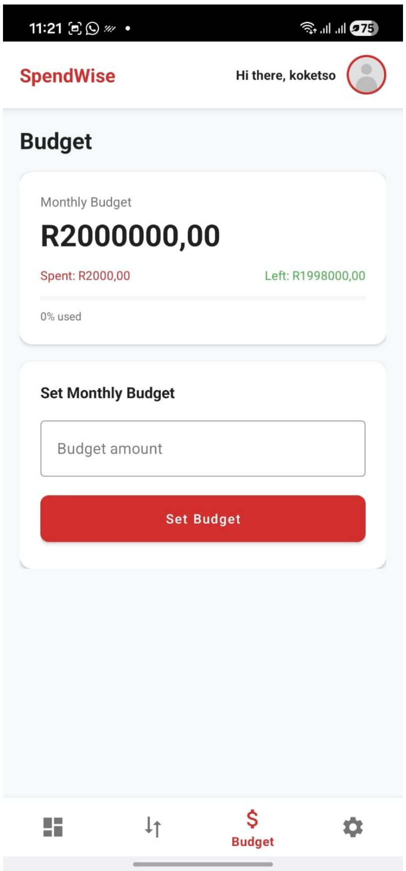


6. **Add/Edit Transaction Modal** → Overlay/popup: Toggle Expense/Income; amount field; description; category dropdown (predefined + custom); date picker; optional photo picker. Cancel/Save buttons. Purpose: Fast, intuitive entry of new data.

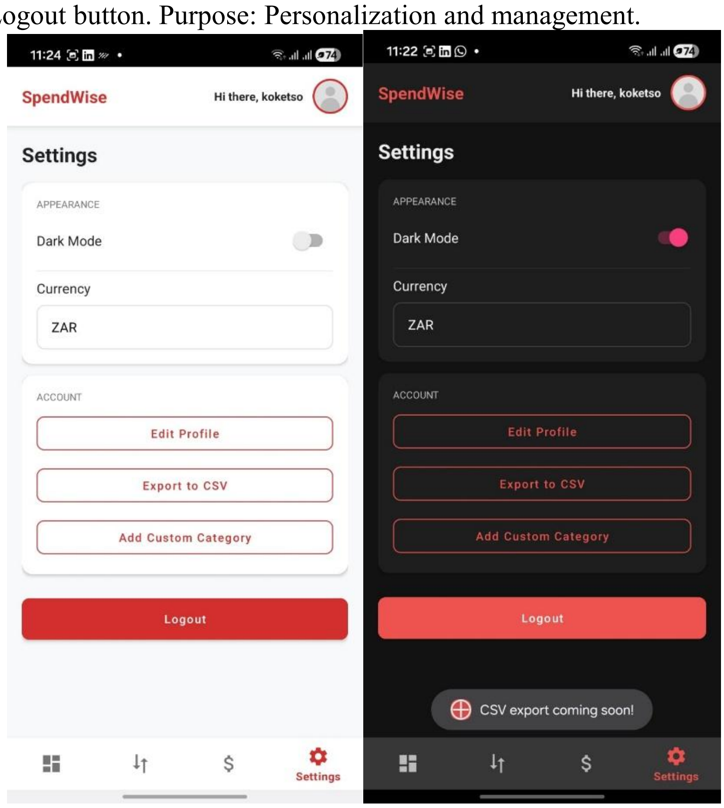




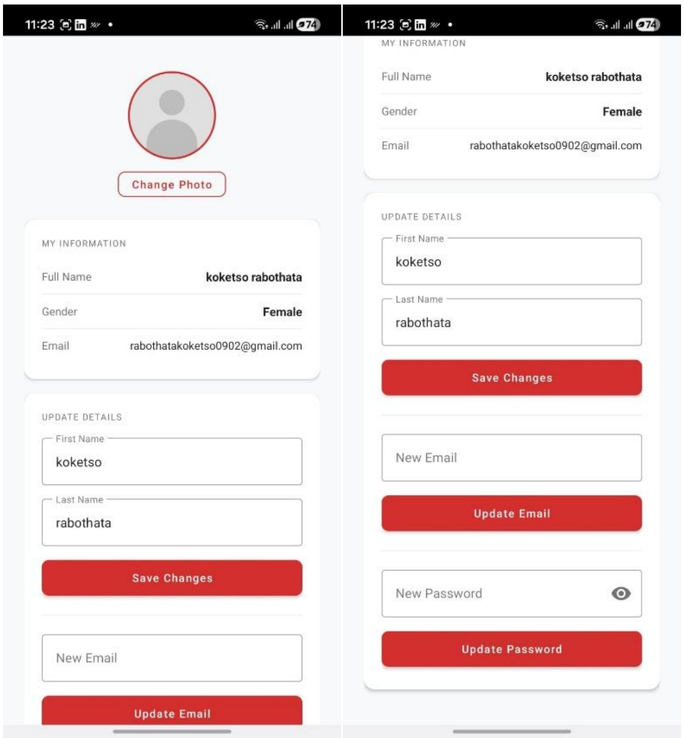
7. **Budget Screen** → Display current monthly budget, spent/left (progress bar, % used); set new budget input + Set button. Purpose: Control and monitor monthly limits.



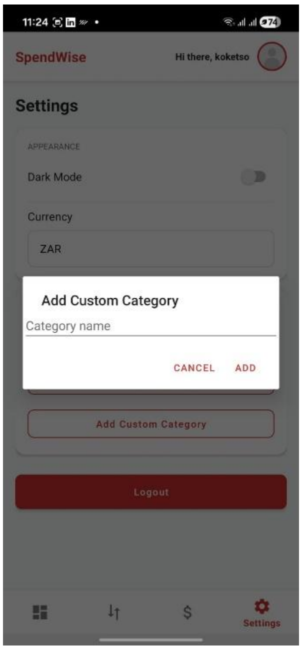
8. **Settings Screen** → Sections: Appearance (dark mode toggle and white mode toggle); Currency (dropdown); Account (Edit Profile, Export CSV placeholder, Add Custom Category). Logout button. Purpose: Personalization and management.



9. **Edit Profile Screen** → View current info (name, gender, email); editable fields (first/last name, new email/password); photo change; Save buttons per section. Purpose: Update personal details.



10. **Add Custom Category Modal** → Input field for name; Cancel/Add. Purpose: Expand categorization flexibility.



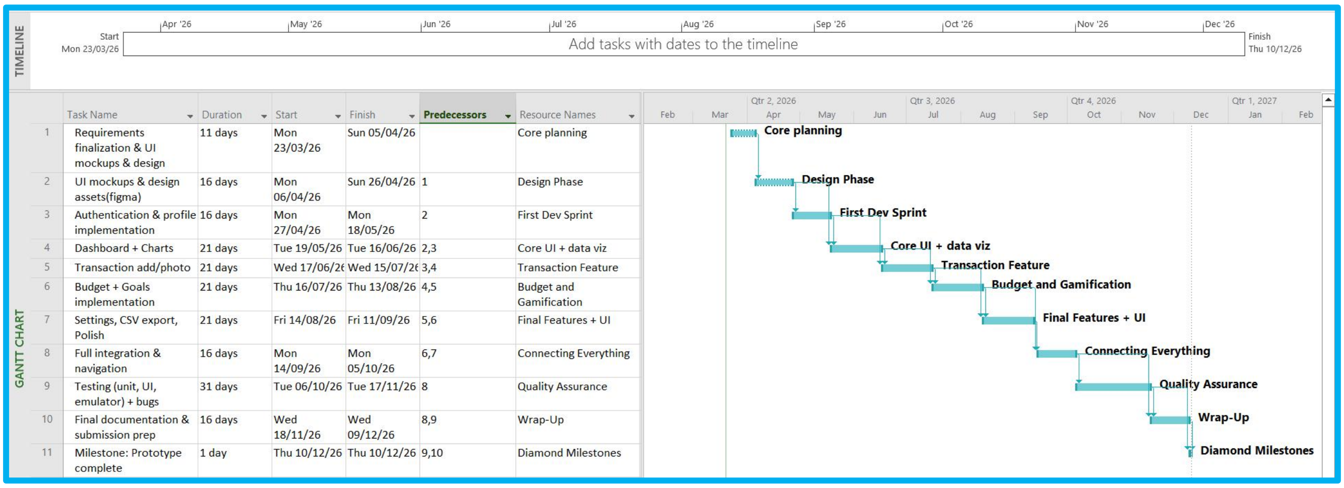
Navigation Diagram :

- Successful login → Dashboard (main app home).
- Register" link → Create Account screen.
- Forgot Password?" → Future password reset flow (placeholder in prototype).

- ✧ Bottom nav bar: Dashboard → Transactions → Budget → Settings (persistent).
- ✧ From Dashboard: Tap + on transactions card? → Transactions screen.
- ✧ From Transactions: Floating + → Add Transaction modal.
- ✧ From Budget: Set Budget button → Input field.
- ✧ From Settings: Currency → Dropdown; Add Custom Category → Modal; Edit Profile → Profile screen.
- ✧ All screens: Back gesture or nav bar switch; logout from Settings → Login.
- ✧ Modals: Dismiss via Cancel or outside tap.

The flow is linear and intuitive: enter data → view summaries → adjust budgets/settings.

5. Project Plan



	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1	Requirements finalization & UI mockups & design	11 days	Mon 23/03/26	Sun 05/04/26		Core planning
2	UI mockups & design assets(figma)	16 days	Mon 06/04/26	Sun 26/04/26	1	Design Phase
3	Authentication & profile implementation	16 days	Mon 27/04/26	Mon 18/05/26	2	First Dev Sprint
4	Dashboard + Charts	21 days	Tue 19/05/26	Tue 16/06/26	2,3	Core UI + data viz
5	Transaction add/photo	21 days	Wed 17/06/26	Wed 15/07/26	3,4	Transaction Feature
6	Budget + Goals implementation	21 days	Thu 16/07/26	Thu 13/08/26	4,5	Budget and Gamification
7	Settings, CSV export, Polish	21 days	Fri 14/08/26	Fri 11/09/26	5,6	Final Features + UI
8	Full integration & navigation	16 days	Mon 14/09/26	Mon 05/10/26	6,7	Connecting Everything
9	Testing (unit, UI, emulator) + bugs	31 days	Tue 06/10/26	Tue 17/11/26	8	Quality Assurance
10	Final documentation & submission prep	16 days	Wed 18/11/26	Wed 09/12/26	8,9	Wrap-Up
11	Milestone: Prototype complete	1 day	Thu 10/12/26	Thu 10/12/26	9,10	Diamond Milestones

6. Conclusion

This document creates a unified, captivating budgeting software based on study findings. Spendwise distinguishes itself by tackling fatigue through gamification, inadequate visualisation with interactive charts and usability through automation and dark mode. Buildable, high-quality implementation that is prepared for prototype testing and iteration is ensured by the comprehensive requirements, mockups, navigation and realistic project plan.

7. References

Bohnenkamp, J., 2024. The impact of digital finance channels on saving and investing behaviour in South Africa. [pdf] South African Reserve Bank. Available at: <https://www.resbank.co.za/content/dam/sarb/what-we-do/payments-and-settlements/cross-border-payments-conference/documents/digital-finance-impact.pdf> [Accessed 20 March 2026].

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